

Virtual Hand Gesture Painting

Satyam V. Tiwari
Department of Information Technology
St. John College of Engineering and Management
Palghar, India
shivaaytiwari55@gmail.com

Vivek R. Verma Department
of Information Technology
St. John College of Engineering and Management
Palghar, India
vivekve2001@gmail.com

Anasali I.Sayyed
Department of Information
Technology
St. John College of Engineering and Management
Palghar, India
anasalisayyed258@gmail.com

Deep A.Vartak
Department of Information Technology
St. John College of Engineering and Management
Palghar, India
deep29vartak@gmail.com

Ms.Soniya Khatu
Department of Information Technology
St. John College of Engineering and Management
Palghar, India
soniya@sjcem.edu.in
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Abstract— In some painting softwares we need to use mouse and keyboard inorder to draw or paint on the screen ,it has some ups and downs and they also contian limitations . In order to overcome this we are developing a virtual hand gesture painting system that uses OpenCV and MediaPipe to enable users to draw shapes and write text in a digital environment. The system will use computer vision algorithms to recognize hand gestures and translate them into digital strokes on a virtual canvas. The system will also allow users to select different colours to create a wide range of shapes and text styles.The project uses the MediaPipe framework to recognize hand gestures and map them to specific actions. The MediaPipe framework is a powerful tool that allows developers to build complex computer vision applications with ease. It includes pre-trained models and algorithms for hand tracking, pose estimation, face detection, and much more. In this project, the hand tracking model is used to recognize and track the position of the user's hand in real-time, while the pose estimation model is used to recognize and interpret the user's hand gestures. Given the real time webcam data, this paint-like python application uses OpenCV library to track an object-of-interest and allows the user to draw by using finger, which makes it both awesome and challenging to draw simple things.

I. INTRODUCTION

Currently, people and machines interact mostly through direct contact methods such as the mouse, keyboard, remote control, touch screen, and other similar devices, whereas people communicate primarily through natural and intuitive non-contact methods such as sound and

physical motions. Many researchers have attempted to help the computer identify other intentions and information using non-contact methods like as voice, facial expressions, physical motions, and gestures. A gesture is the most crucial aspect of mortal language, and gestures also play a significant role in mortal communication. They're considered to be the simplest way for humans and computers to communicate. Sign language recognition, robotics, and other applications fall under the category of gesture recognition. Two methods are typically used for Gesture recognition for HCI applications. The first is based on wearable or direct physical approaches, while the second is based on computer vision and does not require any sensors to be worn. The data-glove, which is made up of sensors to capture hand motion and location, is used in the wearable or direct contact technique. The vision-based technique uses the camera to provide contactless communication between humans and machines. Computer vision cameras are simple to operate and inexpensive. However, due to differences in hand sizes, hand position, hand orientation, lighting conditions, and other factors, this approach has certain limitations. In this paper, we introduce a virtual paint application that uses hand gestures for real-time drawing or sketching on the canvas. Hand gesture-based paint application can be implemented using cameras to capture hand movement. To accomplish activities like as tool selection, writing on the canvas, and clearing the canvas, an intangible interface is created and implemented using vision-based real-time dynamic hand gestures. The images of the hands are taken with the system's web camera and processed in real time with a single-shot detector model and media pipe, allowing the machine to communicate with its user in a fraction of a second...

II. BACKGROUND

A. Python

Python is a computer programming language often used to build websites and software, automate tasks, and conduct data analysis. Python is a general-purpose language, meaning it can be used to create a variety of different programs and isn't specialized for any specific problems. This versatility, along with its beginner-friendliness, has made it one of the most-used programming languages today. Python is often used to develop the back end of a website or application—the parts that a user doesn't see. Python's role in web development can include sending data to and from servers, processing data and communicating with databases, URL routing, and ensuring security. Python offers several frameworks for web development. Commonly used ones include Django and Flask. Some web development jobs that use Python include back-end engineers, full stack engineers, Python developers, software engineers, and DevOps engineers.

B. OpenCV

OpenCV is an open-source software library for computer vision and machine learning. The OpenCV full form is Open Source Computer Vision Library. It was created to provide a shared infrastructure for applications for computer vision and to speed up the use of machine perception in consumer products. OpenCV, as a BSD-licensed software, makes it simple for companies to use and change the code. There are some predefined packages and libraries that make our life simple and OpenCV is one of them. OpenCV was built to provide a common infrastructure for computer vision applications and to accelerate the use of machine perception in the commercial products.

C. MediaPipe

MediaPipe Solutions provides a suite of libraries and tools for you to quickly apply artificial intelligence (AI) and machine learning (ML) techniques in your applications. You can plug these solutions into your applications immediately, customize them to your needs, and use them across multiple development platforms. MediaPipe Solutions is part of the MediaPipe open source project, so you can further customize the solutions code to meet your application needs. The MediaPipe Hand Landmarker task lets you detect the landmarks of the hands in an image. You can use this Task to localize key points of the hands and render visual effects over the hands. This task operates on image data with a machine learning (ML) model as static data or a continuous stream and outputs hand landmarks in image coordinates, hand landmarks in world coordinates and handedness(left/right hand) of multiple detected hands.

In this system, the output is very distorted because while converting the frame to HSV color it fails to filter the impurities (small contour), which results into a non desirable output. The pixel of the output drawing is not upto the mark.

In 2021, Saurabh Saoji *et al.* [2] Suggested a system which uses methods like faster R-CNN, KCF tracking algorithm. Upcoming object detection algorithms such as YOLO v3 can improve fingertip recognition accuracy and speed. Air-writing systems should only obey their master's control gestures and should not be misled by people around. Advancements in Artificial Intelligence will enhance the efficiency of air-writing.

In 2017, Aashni Haria *et al.* [3] Proposed a Image recognition system which facing many challenges like the extraction of invariant features, transition between gestures, automatic filtering and segmentation of features, matching techniques, mixed gestures recognition and complex backgrounds amongst others.

In 2017, Bhumika Nandwana *et al.* [4] outlined the approach which uses dataset gestures to match against the input gesture, where considerably more complex patterns require complex algorithm. Deep learning technique and artificial intelligence techniques to match the interaction gesture in real time with dataset gestures already containing specific postures or gestures. Although this approach can identify a large number of gestures, it has some drawbacks in some cases, such as missing some gestures because of the classification algorithms accuracy contrast.

In 2022, Niharika M. [5] developed the virtual paint application which allows users to draw anything on screen using hand movements. The user can save their completed work or see their drawing process as a replay animation with this application. In this system the output is not fully satisfying because lines are not clear and smooth. Also system does not gives choice to select variety of brush

In 2019, Prajakta Vidhate *et al.* [6] presented a system which is motivated by the importance of real time communication under It is capable of working in the dark, invariant to signer's skin color, clothing, and background lighting conditions. The system is slow in detecting the hand movements therefor it does not provide accurate and fast result. The system must need to improve the design strategy.

In 2022, Sumedh Bansode *et al.* [7] suggested a system whose functionality referred to the set of functions or services that the system equip is to the users while system can operate and perform specific user purposes activity efficiently such as virtual drawing. The application is having high requirements that's the reason sometime computer get freeze. sometimes its hard for Kinect sensor to detect smaller objects like fingers.

III. LITERATURE REVIEW

In 2021, Pranavi Srungarvarapu *et al.* [1] Presented a system in which user can choose from different color to draw whatever the user wants. The morphological operations used were Erosion and Dilatation.

In 2018, Elliot Hu-Au *et al.* [8] suggested a system in which several characteristics of VR provide an opportunity to boost student engagement. As a hands-on, interactive, immersive experience. VR also provides an opportunity for constructivist learning. The project is mainly focused on the VR this makes it very expensive. And it also requires different sensors for work too.

In 2022, Nethravati TL. *et al.* [9] made this project to use AI models for virtual drawing ;allowing for new modes of expression or simply making the artists' life a little easier. AI has artificial neural networks that are trained to mimic and function like biological networks of the brain. In effect, AI assimilates and interprets the visual and sonic information fed to it, draws its own conclusions, and creates abstract visual variations.

In 2015, Akash Gaikwad *et al.* [10] Proposed a system which proves to be the faster in way of recording and accessing our data, compared to current scenario .The data storage, as soft copy makes it easily transferable and occupies much lesser space than the traditional way. This project require projector ,mirror , and color markers and also mobile computing device , which makes it more complex and increases the budget of the project too.

IV. PROPOSED SYSTEM

The proposed system of Virtual Hand gesture painting is a technology, we are using OpenCV and mediaPipe to detect hand and map hand gesture on to the screen.

A. Block Diagram

The block diagram of the proposed system is displayed in Fig. 1.

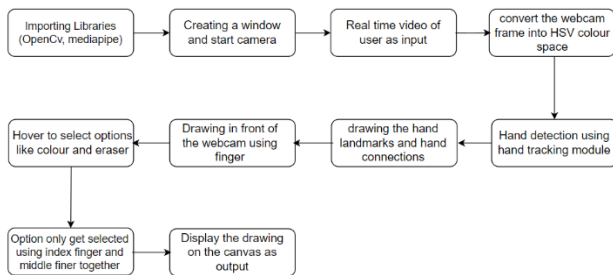


Fig. 1. Block diagram of the proposed system

The aim is to develop virtual paint canvas which is hand-free digital drawing canvas that utilizes OpenCV to recognize and map hand gesture onto a screen. The user can even choose the features such as circle, rectangle etc of the choice that will be displayed. As soon as the user shows up his/her hand in the camera the system detects it. User can select the drawing option by hovering on it. If user shows only index finger then he/she is in the drawing mode. To select different options like brush, shapes and eraser from header and it get selected by using index and middle finger together.

B. Flowchart

The following Figure 2, depicts the flowchart for a Virtual Hand Gesture Painting.

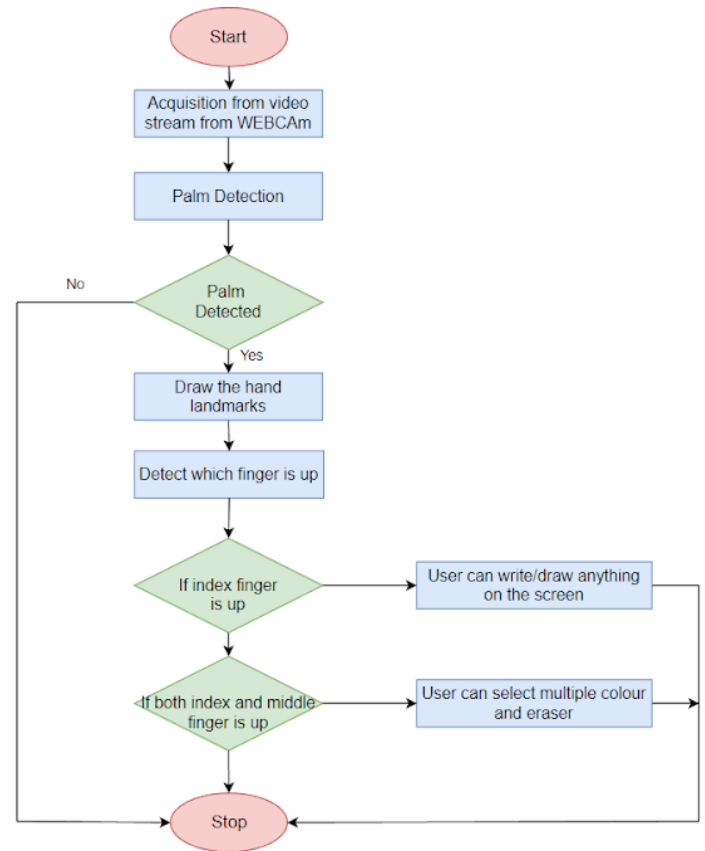


Fig. 2. Flowchart of Virtual Hand Gesture Painting

V. IMPLEMENTATION

Figure 3 shows that, as soon as user show their hand in front of the camera, camera detect his/her hand and display all the landmarks of his/her hand on the screen. There are many colors available on the top of the canvas so that the user can select any color and paint as per his wish.

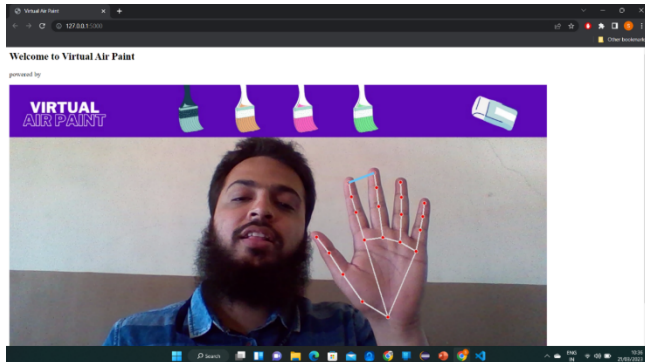


Fig. 3. Home Page

Figure 4 Describes, As you can see the user wants to write (HII) on a screen so he simply draw in the air using its index finger in front of the camera, and camera recognizes and map those hand gesture on screen, as a result the drawing we can see on screen.

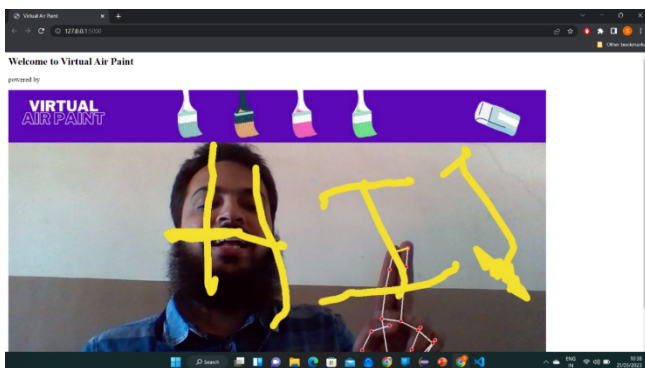


Fig. 4. Drawing on Virtual Board

Figure 5 Shows, Our system also connects to the mobile phone, whatever the user draws in front of the laptop/computer's camera will be visible on the mobile screen as well and the output will be shown.

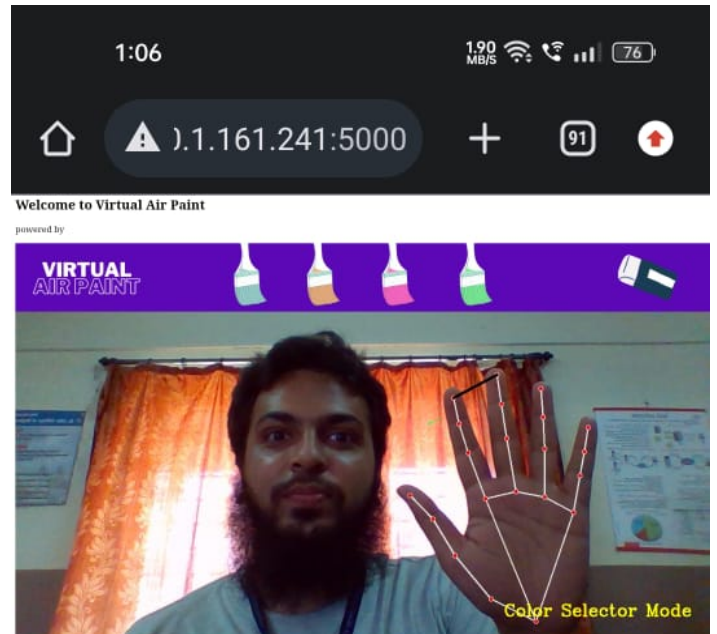


Fig. 5. Connected to Mobile Device

VI. RESULT & DISCUSSIONS

The result of a hand gesture painting project would be a software application or system that allows users to create digital paintings using their hand gestures in a virtual environment. The system would use advanced hand tracking and gesture recognition technology to track the user's hand movements and translate them into brush strokes and other painting actions. Users would be able to select from a variety of painting tools and colors. The success of a hand gesture painting project would depend on a variety of factors, such as the accuracy and responsiveness of the hand tracking and gesture recognition technology, the ease of use and intuitiveness of the user interface, the realism and quality of the rendered artwork, and the scalability and compatibility of the system across different devices and platforms. Overall, the result of a hand gesture painting project would be a compelling and engaging software application that combines the artistry of painting with the interactivity of digital technology, and that provides users with a fun and rewarding experience.

VII. CONCLUSION & FUTURE SCOPE

In conclusion, a virtual hand gesture painting project has the potential to revolutionize the way people create and interact with digital art. By combining the latest advances in hand tracking and gesture recognition technology with a variety of painting tools and a user-friendly interface, this project can provide users with a unique and immersive experience that is both creative and enjoyable. The result of the project would be a software application that allows users to express their creativity and explore their artistic potential in a new and exciting way, while also providing the convenience and versatility of digital technology. Ultimately, a virtual hand gesture painting project has the potential to inspire and engage users of all ages and backgrounds, and to unlock new possibilities in the world of digital art. Overall, Advanced Virtual Paint Application using OpenCV and Hand tracking module, we have achieved a hands-free drawing program that uses OpenCV to detect the user's pointer finger.

Virtual air hand gesture painting could be used in a variety of educational and training contexts, allowing students to learn about art and design in a more interactive and engaging way. For example, students could use this technology to practice drawing and painting techniques in a virtual environment. Virtual air hand gesture painting could be integrated with other technologies, such as virtual reality and augmented reality, to create even more immersive experiences. This could open up new possibilities for gaming, entertainment, and other industries. Overall, the future of virtual air hand gesture painting looks bright, and we can expect to see continued innovation and development in this exciting field.

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